

Input Devices — Detailed Lecture Notes

JKSSB Computer Awareness

October 8, 2026

12.1 What is an input device

A computer works in a cycle: input, process, and output. In the input step, data and instructions are fed into the machine. An **input device** is the hardware that accepts data or control signals from the user or from the environment and passes them to the computer, converting a human action or a physical signal into machine-readable form. Without an input device the computer would have nothing to process.

Definition (Input device). *Hardware that accepts data or control signals from the user or the environment and supplies them to the CPU, converting them into a form the computer can process.*

Common input devices include the keyboard, mouse, touchscreen, scanner, microphone, digital camera, barcode reader, and biometric sensor. The following table fixes the meaning of each term used in this lesson.

Term	Meaning in this lesson
Keyboard	Text input device that converts a keystroke into a character code.
Mouse	Pointing device moved on a surface to control the on-screen pointer.
Trackball	Pointing device whose ball is rotated in place; the device stays still.
Touchpad	Flat, touch-sensitive pad on a laptop; the pointer moves on the screen.
Joystick	Lever that moves on two or three axes, used for games and control.
Light pen	Pen-shaped input device with a light sensor, used on a CRT screen.
Stylus / digitizer	Pen plus tablet that converts strokes into digital coordinates.
Scanner	Converts a paper document or picture into a digital image.
Biometrics	Input from a physical or behavioural human trait (fingerprint, iris, face).

12.2 Classifying input devices

Input devices are grouped by the type of data they capture. This is the classification JKSSB question-setters use most often, so learn one example for each row.

Category	Typical devices
Text input	Keyboard
Pointing input	Mouse, trackball, touchpad, joystick, light pen, stylus, touchscreen
Audio input	Microphone
Image / video input	Scanner, digital camera, webcam
Biometric input	Fingerprint, iris, and face sensors
Code / tag input	Barcode, QR code, RFID

A single device may belong to more than one row if it captures more than one kind of data, but for exam purposes use the primary role: a touchscreen is a pointing input device, a webcam is an image/video input device, and a microphone is an audio input device.

12.3 Direct vs indirect input

A second useful classification is whether the data is captured at its source or encoded by the user.

Direct input devices capture the data themselves. Examples are the touchscreen, light pen, scanner, barcode reader, biometric sensor, and microphone: the device reads the object, code, trait, or sound and turns it directly into computer data.

Indirect input devices require the user to act on the device to encode the data. Examples are the keyboard, mouse, trackball, touchpad, and joystick: the user translates an intention into keystrokes or pointer movements, which the device then sends to the computer.

Definition (Direct input). *Input in which the device captures the data at its source (for example, a scanner reading a page or a biometric sensor reading a fingerprint) without the user encoding it by hand.*

The distinction matters because it explains why some devices are fast and error-free while others depend on the user. A barcode reader or a biometric sensor captures the data exactly as it is, so there is no typing mistake; a keyboard, being indirect, is only as accurate as the person typing. In a question, if the stem stresses that the device *reads* or *captures* an object, code, trait, or sound, think of a direct input device; if it stresses that the *user types or moves* something to enter the data, think of an indirect input device.

12.4 The keyboard: text input

The keyboard is the most common text input device. Its standard letter layout is called **QWERTY**, after the first six letters of the top row. A keyboard has alphanumeric keys, function keys (F1 to F12), modifier keys such as Ctrl, Alt, and Shift, and usually a numeric keypad. Each time a key is pressed, the keyboard sends the corresponding character code to the computer. The keyboard is an indirect input device: the user types the data, and the device encodes it.

Keyboards differ in construction even though their role is the same. A **membrane keyboard** uses a flexible membrane under the keys and is cheap and quiet. A **mechanical keyboard** uses a separate physical switch under each key and gives a firmer, more definite feel. A **wireless keyboard** connects to the computer by Bluetooth or by a USB receiver instead of a cable. Whatever the construction, a keyboard is still a text input device: it sends character codes for the keys that are pressed.

12.5 Pointing devices

A pointing device controls the position of an on-screen pointer (cursor) and is used to point, click, and drag. The table below fixes each pointing device and its nearest neighbour, because JKSSB questions often ask candidates to tell two similar pointing devices apart.

Device	How it works	Nearest neighbour
Mouse	The device slides on a surface and the pointer follows.	Trackball: the ball rotates while the device stays still.
Trackball	The ball is rotated in place; the device is stationary.	Mouse: the whole device moves.
Touchpad	A flat finger pad moves the pointer on the screen (indirect).	Touchscreen: the display itself is touched (direct).
Joystick	A lever moves on two or three axes for games and control.	Mouse: flat-surface pointing.
Light pen	A light sensor is pointed at a CRT screen.	Stylus: used on a digitizer surface.
Stylus + digitizer	A pen stroke is converted into digital coordinates.	Light pen: uses a light sensor, not a tablet.

12.5.1 The mouse

The **mouse** is a hand-held pointing device moved on a flat surface. It has one or more buttons and usually a scroll wheel. An **optical or laser mouse** tracks movement with light, which is the common modern design; an older **mechanical mouse** used a rolling ball. A mouse is an input device (PYQ-08).

12.5.2 The trackball

The **trackball** is, in effect, an upside-down mouse: the ball is on top and is rotated with the fingers while the device itself stays still. This saves desk space because the device does not have to be moved. The nearest-neighbour trap is that a trackball and a mouse both point, but with a mouse the *device* slides, while with a trackball the *ball* rotates in place.

12.5.3 The trackpad or touchpad

The **trackpad** (also called the touchpad) is a flat, touch-sensitive pad found on laptops. The user slides a finger across the pad to move the pointer on the screen and taps to click. It is an indirect device because the finger does not touch the object on the screen; the pad only controls the pointer.

12.5.4 The joystick

The **joystick** is a lever that can be moved on two or three axes. It is used mainly for games and for some industrial control tasks.

12.5.5 The light pen

The **light pen** is a pen-shaped input device with a light sensor at its tip. It is pointed at a CRT screen to select items or to draw. A light pen is an input device, not an output device (TRAP-12).

12.6 Stylus and digitizer tablet

A **stylus** is a pen-like tool used to write or draw on a tablet or a touch screen. A **digitizer** (graphics tablet) is a flat surface that converts the movement of the stylus into digital coordinates. Together they are used for freehand drawing, capturing signatures, computer-aided design (CAD), and handwriting recognition. Because a digitizer captures a drawing as data, it is an input device (TRAP-12). Do not confuse the stylus with the light pen: the light pen works with a light sensor, while the stylus works with a digitizer surface.

12.7 Touchscreen

A **touchscreen** is a display that also works as an input device. The user touches the screen directly to select and enter data. Because the item touched is the item selected, a touchscreen is a direct input device. The two common technologies are **resistive** and **capacitive** touchscreens. Touchscreens are used in smartphones, tablets, ATMs, and information kiosks.

The nearest-neighbour trap is the touchscreen versus the touchpad. On a touchscreen the user touches the display itself, so the input is direct. On a touchpad the user touches a separate pad and the pointer moves on the screen, so the input is indirect. A touchpad is not a touchscreen.

12.8 Scanners, cameras, and webcams

A **scanner** converts a paper document or a picture into a digital image. Common types are the **flatbed** scanner (the page is placed on a glass surface), the sheet-fed scanner, the handheld scanner, and the drum scanner. Software such as OCR (Optical Character Recognition) or OMR (Optical Mark Recognition) can then read text or marks from the scanned image. A scanner is an input device (TRAP-12).

A **digital camera** captures photographs as digital images and stores them on a memory card. A **webcam** is a small digital camera connected to a computer and used mainly for live video, such as video calls and streaming. Both are image/video input devices.

12.9 Barcode, QR code, and RFID

A **barcode** is a one-dimensional pattern of parallel lines of varying width, read by a barcode scanner. It is used on product labels and on book ISBNs.

A **QR code** (Quick Response code) is a two-dimensional matrix of black and white squares. It can store more data than a barcode and can be read from any direction. The full form of QR is **Quick Response**.

RFID stands for **Radio Frequency Identification**. An RFID system has a tag and a reader; the two exchange radio waves, so no line of sight is needed, unlike a barcode. RFID is used in toll collection, access cards, and inventory tracking. A barcode, a QR code, and an RFID tag are all read by an input device that feeds the data into the computer.

12.10 Microphone and biometric input

A **microphone** is an audio input device. It converts sound, such as voice or music, into an electrical signal that the computer digitises. It is used for recording, voice calls, and speech recognition. The matching output device is the **speaker**: a microphone takes audio in, a speaker sends audio out.

Biometric input identifies a person from a physical or behavioural trait. Common traits are the **fingerprint**, the **iris**, the **face**, the voice, and the palm. A sensor captures the trait and feeds the measured data to the computer. Biometrics are used for attendance systems, device unlocking, and authentication. Because the sensor reads a human trait and passes the data to the system, biometric input is a direct input form.

Example 12.10.1. *A student wants to scan a printed page into the computer. The correct device is a scanner, which converts the page into a digital image. If the student instead wanted to type the same text by hand, the device would be a keyboard, which is an indirect text input device. If the goal were to capture a live video of the student, the device would be a webcam.*

12.11 Summary of input devices

The table below gathers the devices in this lesson for quick revision. Use it as a checklist: for each device, be ready to state its category and its one-line function, and to name the device it is most often confused with.

Device	Category	Function
Keyboard	Text	Converts a keystroke into a character code.
Mouse	Pointing	Moves the on-screen pointer; point, click, drag.
Trackball	Pointing	Ball rotated in place; device stays still.
Touchpad	Pointing	Flat finger pad; pointer moves on the screen.
Joystick	Pointing	Lever on two or three axes for games/control.
Light pen	Pointing	Light sensor pointed at a CRT screen.
Stylus + digitizer	Pointing	Converts pen strokes into digital coordinates.
Touchscreen	Pointing (direct)	Touch the display to select and enter.
Scanner	Image	Converts a document or picture into a digital image.
Digital camera	Image/video	Captures photographs as digital images.
Webcam	Image/video	Small camera for live video on a computer.
Microphone	Audio	Converts sound into an electrical signal.
Biometric sensor	Biometric	Reads a fingerprint, iris, or face trait.
Barcode / QR code	Code/tag	Read optical codes that identify an item.
RFID	Code/tag	Reads a tag by radio waves; no line of sight.

12.12 Points of confusion between similar devices

Beyond classifying a device, JKSSB items often test a single distinction between two devices that look or sound alike. The table below collects the pairs used in this lesson.

Pair	The distinction to state
Mouse vs trackball	Mouse: the device slides. Trackball: the ball rotates, device still.
Touchscreen vs touchpad	Touchscreen: touch the display (direct). Touchpad: separate pad moves the pointer (indirect).
Light pen vs stylus	Light pen: light sensor on a CRT screen. Stylus: used with a digitizer tablet.
Scanner vs digital camera	Scanner: converts a flat document or picture into an image. Camera: captures a scene.
Barcode vs QR code	Barcode: 1D parallel lines. QR code: 2D matrix, holds more data.
Barcode/QR vs RFID	Barcode/QR are read optically and need line of sight; RFID uses radio waves.
Microphone vs speaker	Microphone: audio input. Speaker: audio output.
Data glove vs haptic glove	Data glove: captures hand movement (input). Haptic glove: gives force feedback (output).

12.13 Exam retrieval and common confusions

Six distinctions prevent most errors on this topic.

1. A light pen is an **input** device; a plotter is an **output** device. Do not swap them (TRAP-12).
2. A punch-card reader, a digitizer tablet, and a scanner are all **input** devices. A drum printer and a sound card are **output** devices (TRAP-12).
3. A **trackball** rotates its ball in place; a **mouse** slides over a surface.
4. A **touchscreen** is touched directly; a **touchpad** is a separate pad that moves the pointer indirectly.
5. **RFID** uses radio waves and needs no line of sight; a **barcode** and a **QR code** are read optically.
6. **Haptic gloves**, a **tactile display**, and a **Voice Response Unit (VRU)** give feedback to the user and are **output**; a **data glove** captures hand movement and is **input** (PYQ-09).

Example 12.13.1. *An item lists four devices — light pen, punch-card reader, plotter, and scanner — and asks which are input devices. Classify each one: the light pen is input, the punch-card reader is input, the plotter is output, and the scanner is input. The answer is the light pen, the punch-card reader, and the scanner. The plotter is the only output device in the list.*

Exercises

1. Define an input device and give three examples.
2. Classify these devices by the type of data they capture: keyboard, microphone, scanner, fingerprint sensor.
3. Distinguish direct input from indirect input, giving one example of each.

4. Explain the difference between a trackball and a mouse, and between a touchscreen and a touchpad.
5. Differentiate a barcode, a QR code, and RFID.
6. State whether each is input or output: light pen, plotter, digitizer tablet, drum printer, sound card.

Solutions

1. An input device accepts data or control signals from the user or the environment and passes them to the CPU in machine-readable form. Examples: keyboard, mouse, scanner (any three valid devices).
2. Keyboard — text input; microphone — audio input; scanner — image input; fingerprint sensor — biometric input.
3. Direct input captures the data at its source, such as a scanner or a biometric sensor. Indirect input needs the user to encode the data, such as a keyboard or a mouse.
4. With a mouse, the device slides over a surface; with a trackball, the ball is rotated in place and the device stays still. A touchscreen is touched directly on the display; a touchpad is a separate pad that moves the pointer indirectly.
5. A barcode is a 1D pattern of lines read optically; a QR code is a 2D matrix that holds more data; RFID uses radio waves between a tag and a reader and needs no line of sight.
6. Light pen — input; plotter — output; digitizer tablet — input; drum printer — output; sound card — output.